

HARSH BAFNA

Portfolio Website

PROFESSIONAL SUMMARY

Machine Learning Engineer with ~3 years of experience specializing in Generative AI, multi-agent system orchestration, and production-grade LLM applications. Proven track record of architecting scalable retrieval (RAG) systems and leading engineering teams to deploy high-impact AI solutions.

EXPERIENCE

Machine Learning Engineer <i>Quantiphi</i> <i>RCA, Segmentation & NL2SQL Bot</i>	Bengaluru, IN <i>Feb 2024 – Present</i> <i>Python, SQL, ADK, LLMs, AI Agents, Multi-Agent Orchestration</i>
- Led a team of 5+ ML engineers within a 24+ member team to design and ship 3 production-grade multi-agent orchestration systems, enabling non-technical users and C-suite executives to query enterprise data through natural-language chat. - Architected and deployed a multi-agent analytics platform on Google's ADK framework with a custom RAG-based retrieval layer over table schemas, spanning 14+ structured data sources (Kafka topics, Google Ads, Google Analytics) to power accurate, context-aware NL2SQL generation. - Cut executive data-reporting and analysis turnaround from days to minutes, accelerating data-driven decision-making across the organization.	
AI Search & Recommendation Engine	<i>Python, Vertex AI Search, LLMs, Ranking</i>
- Designed and architected a conversational search and recommendation engine, replacing traditional keyword-based search with natural-language query understanding. - Engineered a conversational search platform leveraging Vertex AI Search for Commerce, indexing an 80K+ product SKU catalog, alongside recommendation models (recommended-for-you, similar items, frequently-bought-together) to surface more relevant results. - Elevated search from keyword/fuzzy matching to free-form conversational queries, improving product-catalog sorting accuracy against user intent.	
AI Email Campaign Generator	<i>Python, LLMs, AI Workflow, Web-scraping, NLP, Imagen</i>
- Architected an AI-driven email campaign generator that scrapes theme and metadata from property websites and applies it to auto-generated marketing emails. - Built scheduled extraction jobs to pull property metadata/themes and an AI workflow to generate each section of the email, including image enhancement via Imagen models. - Cut email-campaign drafting time for analysts from days/weeks to minutes by automating theme extraction and end-to-end content generation.	
Lead Hotspot Detection	<i>Python, LangChain, LLMs</i>
- Developed and evaluated multiple ML models to predict lead-contamination hotspots, selecting the best-performing hybrid approach. - Cleaned and engineered features from PWS data sourced from the SDWIS government database, performing EDA and combining model-driven predictions with rule-based decision logic. - Achieved 88+% accuracy in lead hotspot detection, driving more effective resource allocation for remediation efforts.	
Software Engineer Intern 	Remote
<i>JP Morgan Chase & Co.</i>	<i>Nov 2022 – Sept 2023</i>
- Designed a real-time data streaming interface capable of ingesting and processing over 100,000 data points per second. - Utilized JP Morgan Chase's Perspective open-source data visualization tool to build a live graph showcasing the historical correlation between two stocks. - Implemented upper and lower-bound indicators based on correlation analysis, enabling real-time monitoring of key financial metrics.	

EDUCATION

B.M.S. College of Engineering <i>Bachelor of Engineering – Electronics & Instrumentation Engineering; GPA: 8.3/10</i>	Bangalore, India <i>2020 – 2024</i>
S.Tech IT School <i>Higher Secondary School Certificate – PCM; Percentage: 82.6%</i>	Rajasthan, India <i>2019 – 2020</i>

TECHNICAL SKILLS

Languages & Frameworks: Python, SQL, PyTorch, TensorFlow, Scikit-Learn, Keras, Pandas, NumPy, OpenCV
Generative AI & LLMs: LangChain, LangGraph, Google ADK, Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), Prompt Engineering, Embeddings, AI Agents, Multi-Agent Systems, Fine-Tuning, Retrieval Pipelines, Hybrid Search, Reranking
AI & Machine Learning: Deep Learning, Generative AI, Large Language Models (LLMs), Agentic AI, NLP, Computer Vision, Transformers, XGBoost, LightGBM, Hyperparameter Optimization, Document AI (OCR, Extraction), Hugging Face
Data Engineering & Databases: Snowflake, Kafka, BigQuery, PostgreSQL, MongoDB, MySQL, Redis, Elasticsearch, Vector Databases, Data Pipelines, Exploratory Data Analysis (EDA), Time-Series Forecasting
Cloud, Infrastructure & Deployment: Git, GitHub, GCP (Vertex AI), AWS, Azure ML, Docker, Kubernetes, CI/CD, GitHub Actions, Linux, Shell Scripting, FastAPI, Flask, REST APIs, Streamlit

CERTIFICATIONS AND ACHIEVEMENTS

Associate Cloud Engineer, Google Cloud 
One for All, Quantiphi 
Think Tank, Quantiphi 
Unsung hero - Consider It Done, Quantiphi 